

Package ‘scf’

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Title Analyzing the Survey of Consumer Finances

Version 1.0.7

Description Analyze public-use micro data from the Survey of Consumer Finances. Provides tools to download prepared data files, construct replicate-weighted multiply imputed survey designs, compute descriptive statistics and model estimates, and produce plots and tables. Methods follow design-based inference for complex surveys and pooling across multiple imputations. See the package website and the code book for background.

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URL <https://github.com/jncohen/scf>

BugReports <https://github.com/jncohen/scf/issues>

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scf

Analyzing Survey of Consumer Finances Public-Use Microdata

Description

This package provides functions to analyze the U.S. Federal Reserve’s Survey of Consumer Finances (SCF) public-use microdata. It encapsulates the SCF’s multiply-imputed, replicate-weighted structure in a custom object class (`scf_mi_survey`) and supports estimation of population-level statistics, including univariate and bivariate distributions, hypothesis tests, data visualizations, and regression models.

Designed for generalist analysts, the package assumes familiarity with standard statistical methods but not with the complexities of multiply-imputed or survey-weighted data. All functions prioritize transparency, reproducibility, and pedagogical clarity.

Methodological Background

The SCF is one of the most detailed and methodologically rigorous sources of data on U.S. household finances. It is nationally representative, includes an oversample of high-wealth households and households in predominantly Black communities, and provides multiply-imputed estimates for item nonresponse. These features increase the analytical value of the data set but also introduce methodological complexity. Valid inference requires attention to:

- **Survey Weights:** The SCF employs a dual-frame, stratified, and clustered probability sample. Analysts must apply the provided sampling weights to produce population-representative estimates.
- **Replicate Weights:** Each observation is associated with 999 replicate weights, generated using a custom replication method developed by the Federal Reserve. These are used to estimate sampling variance.
- **Multiple Imputation:** The SCF uses multiple imputation to address item nonresponse, providing five imputates per household. Estimates must be pooled across imputates to obtain valid point estimates and standard errors.

The `scf` package provides a structured, user-friendly interface for handling these design complexities, enabling applied researchers and generalist analysts to conduct principled and reproducible analysis of SCF microdata using familiar statistical workflows.

Package Architecture and Workflow

This section recommends a sequence of operations enacted through the package's functions. For an in-depth discussion of the methodological considerations involved in these functions' formulation, see Cohen (2026).

1. **Data Acquisition:** Download the data from Federal Reserve servers to your working directory using `scf_download()`.
2. **Data Loading:** Load the data into R using `scf_load()`. This function returns an `scf_mi_survey` object (described below).
3. **Data Wrangling:** Use `scf_update()` to modify the data, or `scf_subset()` to filter it. These functions return new `scf_mi_survey` objects.
4. **Descriptive Statistics:** Compute univariate and bivariate statistics using functions like `scf_mean()`, `scf_median()`, `scf_percentile()`, `scf_freq()`, `scf_xtab()`, and `scf_corr()`.
5. **Basic Inferential Tests:** Conduct hypothesis tests using `scf_ttest()` for means and `scf_prop_test()` for proportions.
6. **Regression Modeling:** Fit regression models using `scf_ols()` for linear regression, `scf_logit()` for logistic regression, and `scf_glm()` for generalized linear models.
7. **Data Visualization:** Create informative visualizations using `scf_plot_dist()` for distributions, `scf_plot_cbar()` and `scf_plot_bbar()` for categorical data, `scf_plot_smooth()` for smoothers, and `scf_plot_hex()` for hexbin plots.
8. **Diagnostics and Infrastructure:** Use `scf_MIcombine()` to pool results across imputates.

Core Data Object and Its Structure

This suite of functions operate from a custom object class, `scf_mi_survey`, which is created by `scf_design()` via `scf_load()`. Specifically, the object is a structured list containing the elements:

- `mi_design`: A list of five `survey::svrepdesign()` objects (one per imPLICATE)
- `year`: Year of survey
- `n_households`: Estimated number of U.S. households in that year, per data from the Federal Reserve Economic Data (FRED) series TTLHH, accessed 6/17/2025.

Imputed Missing Data

The SCF addresses item nonresponse using multiple imputation (see Kennickell 1998). This procedure generates five completed data sets, each containing distinct but plausible values for the missing entries. The method applies a predictive model to the observed data, simulates variation in both model parameters and residuals, and generates five independent estimates for each missing value. These completed data sets—called *implicates*—reflect both observed relationships and the uncertainty in estimating them. See `scf_MIcombine()` for details.

Mock Data for Testing

A mock SCF dataset (`scf2022_mock_raw.rds`) is bundled in `inst/extdata/` for internal testing purposes. It is a structurally valid `scf_mi_survey` object created by retaining only the first ~200 rows per implicate and only variables used in examples and tests.

This object is intended solely for package development and documentation rendering. It is **not suitable for analytical use or valid statistical inference**.

Theming and Visual Style

All built-in graphics follow a common aesthetic set by `scf_theme()`. Users may modify the default theme by calling `scf_theme()` explicitly within their scripts. See `scf_theme()` documentation for customization options.

Pedagogical Design

The package is designed to support instruction in advanced methods courses on complex survey analysis and missing data. It promotes pedagogical transparency through several features:

- Each implicate’s design object is accessible via `scf_mi_survey$mi_design[[i]]`
- Raw implicate-level data can be viewed directly through `scf_mi_survey$mi_design[[i]]$variables`
- Users can execute analyses on individual implicates or combine them using Rubin’s Rules
- Key functions implement design-based estimation strategies explicitly, such as replicate-weight variance estimation
- Minimal abstraction is used, so each step remains visible and tractable

These features allow instructors to demonstrate how survey weights, replicate designs, and multiple imputation contribute to final results. Students can follow the full analytic path from raw inputs to pooled estimates using transparent, inspectable code and data structures.

Author(s)

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References

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Lumley T. Analysis of complex survey samples. doi:10.18637/jss.v009.i08.

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U.S. Federal Reserve. Codebook for 2022 Survey of Consumer Finances. <https://www.federalreserve.gov/econres/scfind>

See Also

Useful links:

- <https://github.com/jncohen/scf>
- Report bugs at <https://github.com/jncohen/scf/issues>

scf_activate_theme	<i>Default Plot Theme for SCF Visualizations</i>
--------------------	--

Description

The theme is designed to:

- Render cleanly in **print** (single-column or wrapped layout)
- Scale well on **HD desktop** monitors without visual clutter
- Remain **legible on mobile** with clear fonts and sufficient contrast

The default figure dimensions assumed for export are **5.5 inches by 5.5 inches** at **300 dpi**, which balances compactness with accessibility across media.

All theme settings are exposed via comments to enable easy brand customization.

Usage

```
scf_activate_theme()
```

```
scf_theme(base_size = 13, base_family = "sans", grid = TRUE, axis = TRUE, ...)
```

Arguments

base_size	Base font size. Defaults to 13.
base_family	Font family. Defaults to "sans".
grid	Logical. Show gridlines? Defaults to TRUE.
axis	Logical. Include axis ticks and lines? Defaults to TRUE.
...	Additional arguments passed to <code>ggplot2::theme_minimal()</code> .

Details

Defines the SCF package's default ggplot2 theme, optimized for legibility, clarity, and aesthetic coherence across print, desktop, and mobile platforms.

Value

`scf_theme()` returns a `ggplot2` theme object. `scf_activate_theme()` returns `NULL` invisibly; called for its side effect of setting the session-wide `ggplot2` theme.

Global activation

`scf_activate_theme()` sets `scf_theme()` as the default `ggplot2` theme for all plots in the current R session — equivalent to `ggplot2::theme_set(scf_theme())`. The effect lasts only for the session and does not persist across sessions. All `scf_plot_*`() functions apply `scf_theme()` internally, so this is most useful when producing custom plots alongside the package's built-in visualizations.

See Also

[ggplot2::theme\(\)](#), [scf_plot_dist\(\)](#)

Examples

```
library(ggplot2)
ggplot(mtcars, aes(factor(cyl))) +
  geom_bar(fill = "#0072B2") +
  scf_theme()

# Activate globally for the session:
scf_activate_theme()
```

scf_corr

Estimate Correlation Between Two Continuous Variables in SCF Microdata

Description

This function estimates the linear association between two continuous variables using Pearson's correlation. Estimates are computed within each implicate and then pooled across implicates to account for imputation uncertainty.

Usage

```
scf_corr(scf, var1, var2)

## S3 method for class 'scf_corr'
summary(object, ...)
```

Arguments

<code>scf</code>	An <code>scf_mi_survey</code> object, created by scf_load()
<code>var1</code>	One-sided formula specifying the first variable
<code>var2</code>	One-sided formula specifying the second variable
<code>object</code>	A <code>scf_corr</code> object returned by scf_corr() .
<code>...</code>	Currently unused; included for S3 generic compatibility.

Details

Computes the Pearson correlation coefficient between two continuous variables using multiply-imputed, replicate-weighted SCF data. Returns pooled estimates and standard errors using Rubin's Rules.

Value

An object of class `scf_corr`, containing:

results Data frame with pooled correlation estimate, standard error, t-statistic, degrees of freedom, p-value, and minimum/maximum values across implicates.

imps Named vector of implicate-level correlations.

aux Variable names used in the estimation.

Implementation

- Inputs: an `scf_mi_survey` object and two one-sided formulas (e.g., `~income`)
- Correlation computed using `cor(..., use = "complete.obs")` within each implicate
- Rubin's Rules applied to pool results across implicates

Interpretation

Pearson's r ranges from -1 to +1 and reflects the strength and direction of a linear bivariate association between two continuous variables. Values near 0 indicate weak linear association. Note that the operation is sensitive to outliers and does not capture nonlinear relationships nor adjust for covariates.

Statistical Notes

Correlation is computed within each implicate using complete cases. Rubin's Rules are applied manually to pool estimates and calculate total variance. This function does not use `scf_MIcombine()`, which is intended for vector-valued estimates; direct pooling is more appropriate for scalar statistics like correlation coefficients.

Note

Degrees of freedom are approximated using a simplified Barnard–Rubin adjustment, since correlation is a scalar quantity. Interpret cautiously with few implicates.

See Also

`scf_plot_hex()`, `scf_ols()`

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("corr_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)
```

```
# Example for real analysis: Correlate income and net worth
corr <- scf_corr(scf2022, ~income, ~networth)
print(corr)
summary(corr)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_deflate

Convert SCF Dollar Estimates to Real Terms

Description

Multiplies dollar-valued point estimates and their standard errors by the CPI-U-RS deflation factor $\text{CPI}[\text{base_year}] / \text{CPI}[\text{survey_year}]$, converting nominal survey-year dollars to real dollars of the chosen base year. The CPI-U-RS September values are taken directly from the Federal Reserve Board's SCF Bulletin SAS macro.

Usage

```
scf_deflate(x, base_year = 2022)
```

Arguments

x	An object of class <code>scf_mean</code> , <code>scf_median</code> , <code>scf_percentile</code> , or <code>scf_ttest</code> .
base_year	Integer. Reference year for real dollars. Must be a valid SCF survey year (1989–2022, triennial). Default is 2022.

Details

Standard errors rescale correctly under linear multiplication, so both estimates and SEs are multiplied by the same factor. Confidence intervals and group means are also rescaled. The t-statistic, degrees of freedom, and p-value in `scf_ttest` results are invariant to this rescaling and are left unchanged.

Supported functions: `scf_mean`, `scf_median`, `scf_percentile`, and `scf_ttest` return dollar-valued estimates that transform correctly under scalar multiplication. `scf_freq`, `scf_xtab`, and `scf_prop_test` return proportions and are not supported. `scf_corr` returns a dimensionless coefficient and is not supported. The regression functions (`scf_ols`, `scf_glm`, `scf_logit`, `scf_quantreg`) are not supported because deflating coefficients post hoc is ambiguous when a model mixes dollar and non-dollar variables; for real-dollar regression results, deflate variables upstream with `scf_update()` before fitting.

Income note: SCF income is measured in the prior calendar year, so a mean income estimate from the 2019 survey is in 2018 dollars, not 2019 dollars. `scf_deflate()` applies $\text{CPI}[\text{base}] / \text{CPI}[2019]$, which is a close but not exact conversion for income. This is documented in the Federal Reserve's SAS macro, which applies a separate lag adjustment to income before the main deflation step.

Value

The input object with dollar estimates, standard errors, and confidence intervals rescaled to `base_year` dollars. Attributes `"deflated"` (logical) and `"base_year"` (integer) are set on the returned object.

Conditions

Warning — double deflation If the input object has already been deflated (`attr(x, "deflated")` is `TRUE`), `scf_deflate()` issues a warning and proceeds. Applying the function twice compounds the CPI adjustment and produces incorrect results. Check `attr(result, "deflated")` and `attr(result, "base_year")` before calling.

Error — stale result object If the result object does not carry a survey year (`x$aux$year` is `NULL`), the function stops with a message asking you to re-run the originating function (`scf_mean()`, `scf_median()`, etc.) under the current package version. This field was introduced in version 1.0.6.

Error — unsupported class Passing an object of an unsupported class (e.g., a regression result or a frequency table) stops with a message listing the supported classes: `scf_mean`, `scf_median`, `scf_percentile`, and `scf_ttest`.

Error — year not in CPI table If `base_year` is not one of the valid triennial SCF survey years (1989–2022), the function stops and lists the valid options.

See Also

[scf_mean\(\)](#), [scf_median\(\)](#), [scf_percentile\(\)](#), [scf_ttest\(\)](#), [scf_update\(\)](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("deflate_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Deflate a mean estimate to real 2022 dollars
result <- scf_mean(scf2022, ~networth)
result_real <- scf_deflate(result, base_year = 2022)

# Works with median and percentile results
med <- scf_median(scf2022, ~networth)
med_real <- scf_deflate(med, base_year = 2022)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

Description

Wraps a list of replicate-weighted survey designs into an `scf_mi_survey` object. This is called internally by [scf_load\(\)](#), but is also available directly for users who construct their own implicate-level designs outside the standard download-and-load workflow — for example, when integrating external or custom-prepared SCF data files.

Each element of design must be a [survey::svrepdesign\(\)](#) object representing one SCF implicate with replicate weights.

Usage

```
scf_design(design, year, n_households)
```

Arguments

<code>design</code>	A list of five survey::svrepdesign() objects (one per implicate).
<code>year</code>	Numeric SCF survey year (e.g., 2022).
<code>n_households</code>	Numeric total U.S. households represented in year.

Value

An object of class "scf_mi_survey" with:

mi_design List of replicate-weighted designs (one per implicate).

year SCF survey year.

n_households Estimated number of U.S. households.

See Also

[scf_load\(\)](#), [scf_update\(\)](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("design_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Construct scf_mi_survey object
obj <- scf_design(
  design = scf2022$mi_design,
  year = 2022,
  n_households = attr(scf2022, "n_households")
)
class(obj)
length(obj$mi_design)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_download

*Download and Prepare SCF Microdata for Local Analysis***Description**

Downloads SCF public-use microdata from official servers. For each year, this function retrieves five implicates, merges them with replicate weights and official summary variables, and saves them as .rds files ready for use with `scf_load()`.

Usage

```
scf_download(years = seq(1989, 2022, 3), overwrite = FALSE, verbose = TRUE)
```

Arguments

years	Integer vector of SCF years to download (e.g., <code>c(2016, 2019)</code>). Must be triennial from 1989 to 2022.
overwrite	Logical. If TRUE, re-download and overwrite existing .rds files. Default is FALSE.
verbose	Logical. If TRUE, display progress messages. Default is TRUE.

Value

A character vector of paths to the .rds files written to disk (one per year). Each file contains a list of five implicate data frames ready for use with `scf_load()`.

Implementation

This function downloads from official servers three types of files for each year:

- five versions of the dataset (one per implicate), each stored as a separate data frame in a list
- a table of replicate weights, and
- a data table with official derivative variables

These tables are collected to a list and saved to an .rds format file in the working directory. By default, the function downloads all available years.

Details

The SCF employs multiply-imputed data sets to address unit-level missing data. Each household appears in one of five implicates. This function ensures all implicates are downloaded, merged, and prepared for downstream analysis using `scf_load()`, `scf_design()`, and the `scf` workflow.

References

U.S. Federal Reserve. Codebook for 2022 Survey of Consumer Finances. <https://www.federalreserve.gov/econres/scfindo>

See Also

`scf_load()`, `scf_design()`, `scf_update()`

Examples

```
if (FALSE) {
  # Download and prepare SCF data for 2022
  td <- tempfile("download_")
  dir.create(td)

  old <- getwd()
  setwd(td)
  scf_download(2022)

  # Load into a survey design object
  scf2022 <- scf_load(2022, data_directory = td)

  # Cleanup for package check
  unlink(td, recursive = TRUE, force = TRUE)
  setwd(old)
}
```

scf_freq

Estimate the Frequencies of a Discrete Variable from SCF Microdata

Description

This function estimates the relative frequency (proportion) of each category in a discrete variable from the SCF public-use microdata. Use this function to discern the univariate distribution of a discrete variable.

Usage

```
scf_freq(scf, var, by = NULL, percent = TRUE)
```

Arguments

scf	A <code>scf_mi_survey</code> object created by <code>scf_load()</code> . Must contain five replicate-weighted implicates.
var	A one-sided formula specifying a categorical variable (e.g., <code>~racecl</code>).
by	Optional one-sided formula specifying a discrete grouping variable (e.g., <code>~own</code>).
percent	Logical. If TRUE (default), scales results and standard errors to percentages.

Details

Computes weighted proportions and standard errors for a discrete variable in multiply-imputed SCF data, optionally stratified by a grouping variable. Proportions and standard errors are computed separately within each implicate using `svymean()`, then averaged across implicates using SCF-recommended pooling logic. Group-wise frequencies are supported, but users may find the features of `scf_xtab()` to be more useful.

Value

A list of class "scf_freq" with:

results Pooled category proportions and standard errors, by group if specified.

imps A named list of implicate-level proportion estimates.

aux Metadata about the variable and grouping structure.

Details

Proportions are estimated within each implicate using `survey::svymean()`, then pooled using the standard MI formula for proportions. When a grouping variable is provided via `by`, estimates are produced separately for each group-category combination. Results may be scaled to percentages using the `percent` argument.

Estimates are pooled using the standard formula:

- The mean of implicate-level proportions is the point estimate
- The standard error reflects both within-implicate variance and across-implicate variation

Unlike means or model parameters, category proportions do not use Rubin's full combination rules (e.g., degrees of freedom).

See Also

[scf_xtab\(\)](#), [scf_plot_dist\(\)](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("freq_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Proportions of homeownership
scf_freq(scf2022, ~own)

# Example for real analysis: Homeownership proportions by education
scf_freq(scf2022, ~own, by = ~edcl)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_glm

Estimate Generalized Linear Model from SCF Microdata

Description

Estimates generalized linear models (GLMs) with SCF public-use microdata. Use this function when modeling outcomes that follow non-Gaussian distributions (e.g., binary or count data). Rubin's Rules are used to combine implicate-level coefficient and variance estimates.

GLMs are performed across SCF implicates using `svyglm()` and returns pooled coefficients, standard errors, z-values, p-values, and fit diagnostics including AIC and pseudo-R-Squared when applicable.

Usage

```
scf_glm(object, formula, family = binomial())
```

Arguments

<code>object</code>	A <code>scf_mi_survey</code> object, typically created using <code>scf_load()</code> and <code>scf_design()</code> .
<code>formula</code>	A valid model formula, e.g., <code>rich ~ age + factor(edcl)</code> .
<code>family</code>	A GLM family object such as <code>binomial()</code> , <code>poisson()</code> , or <code>gaussian()</code> . Defaults to <code>binomial()</code> .

Value

An object of class "scf_glm" and "scf_model_result" with:

results A data frame of pooled coefficients, standard errors, z-values, p-values, and significance stars.

fit A list of fit diagnostics including mean and SD of AIC; for binomial models, pseudo-R2 and its SD.

models A list of implicate-level `svyglm` model objects.

call The matched function call.

Implementation

This function fits a GLM to each implicate in a `scf_mi_survey` object using `survey::svyglm()`. The user specifies a model formula and a valid GLM family (e.g., `binomial()`, `poisson()`, `gaussian()`). Coefficients and variance-covariance matrices are extracted from each implicate and pooled using Rubin's Rules.

Details

Generalized linear models (GLMs) extend linear regression to accommodate non-Gaussian outcome distributions. The choice of family determines the link function and error distribution. For example:

- `binomial()` fits logistic regression for binary outcomes
- `poisson()` models count data
- `gaussian()` recovers standard OLS behavior

Model estimation is performed independently on each implicate using `svyglm()` with replicate weights. Rubin's Rules are used to pool coefficient estimates and variance matrices. For the pooling procedure, see `scf_MIcombine()`.

Internal Suppression

For CRAN compliance and to prevent diagnostic overload during package checks, this function internally wraps each implicate-level model call in `suppressWarnings()`. This suppresses the known benign warning:

```
"non-integer #successes in a binomial glm!"
```

which arises from using replicate weights with `family = binomial()`. This suppression does not affect model validity or inference. Users wishing to inspect warnings can run `survey::svyglm()` directly on individual implicates via `model$models[[i]]`.

For further background, see: <https://stackoverflow.com/questions/12953045/warning-non-integer-successes-in-a-binomial-glm-survey-packages>

See Also

`scf_ols()`, `scf_logit()`, `scf_regtable()`

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("glm_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Run logistic regression
model <- scf_glm(scf2022, own ~ age + factor(edcl), family = binomial())
summary(model)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_implicates

Extract Implicate-Level Estimates from SCF Results

Description

Returns implicate-level outputs from SCF result objects produced by functions in the `scf` suite. Supports result objects containing implicate-level data frames, `svyestat` summaries, or `svyglm` model fits.

Usage

```
scf_implicates(x, long = FALSE)
```

Arguments

- x** A result object containing implicate-level estimates. Three types are supported:
- Data frame results** Objects from `scf_freq()`, `scf_mean()`, `scf_median()`, `scf_percentile()`, `scf_corr()`, and `scf_xtab()`, whose `$imps` slot contains a list of data frames.
 - Survey statistic results** Objects whose `$imps` slot contains `svyestat` objects (e.g., raw outputs from `survey::svymean()`).
 - Model results** Objects from `scf_ols()`, `scf_glm()`, `scf_logit()`, and `scf_quantreg()`, whose `$imps` or `$models` slot contains `svyglm` fits.
- long** Logical. If `TRUE`, returns stacked data frame. If `FALSE`, returns list.

Value

A list of implicate-level data frames, or a single stacked data frame if `long = TRUE`.

Usage

This function allows users to inspect how estimates vary across the SCF's five implicates, which is important for diagnostics, robustness checks, and transparent reporting.

For example:

```
scf_implicates(scf_mean(scf2022, ~income))
scf_implicates(scf_ols(scf2022, networth ~ age + income), long = TRUE)
```

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("implicates_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Extract implicate-level results
out <- scf_freq(scf2022, ~own)
scf_implicates(out, long = TRUE)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_load

Load SCF Data as Multiply-Imputed Survey Designs

Description

Converts SCF `.rds` files prepared by `scf_download()` into `scf_mi_survey` objects. Each object wraps five implicates per year in replicate-weighted, multiply-imputed survey designs suitable for use with `scf_` functions.

Usage

```
scf_load(min_year, max_year = min_year, data_directory = ".")
```

Arguments

`min_year` Integer. First SCF year to load (1989–2022, divisible by 3).
`max_year` Integer. Last SCF year to load. Defaults to `min_year`.
`data_directory` Character. Directory containing `.rds` files or a full path to a single `.rds` file. Defaults to the current working directory `"."`. For examples and tests, use `tempdir()` to avoid leaving files behind.

Value

Invisibly returns a `scf_mi_survey` (or named list if multiple years). Attributes: `mock` (logical), `year`, `n_households`.

Implementation

Provide a year or range and either (1) a directory containing `scf<year>.rds` files, or (2) a full path to a single `.rds` file. Files must contain five implicate data frames with columns `wgt` and `wt1b1..wt1bK` (typically `K=999`).

See Also

[scf_download\(\)](#), [scf_design\(\)](#), [scf_update\(\)](#), [survey::svrepdesign\(\)](#)

Examples

```
# Using with CRAN-compliant mock data:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("load_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_logit

Estimate Logistic Regression Model using SCF Microdata

Description

Fits a replicate-weighted logistic regression model to multiply-imputed SCF data, returning pooled coefficients or odds ratios with model diagnostics. Use this function to model a binary variable as a function of predictors.

Usage

```
scf_logit(object, formula, odds = TRUE, ...)
```

Arguments

<code>object</code>	A <code>scf_mi_survey</code> object created with <code>scf_load()</code> and <code>scf_design()</code> .
<code>formula</code>	A model formula specifying a binary outcome and predictors, e.g., <code>rich ~ age + factor(edc1)</code> .
<code>odds</code>	Logical. If TRUE (default), exponentiates coefficient estimates to produce odds ratios for interpretability.
<code>...</code>	Additional arguments passed to <code>scf_glm()</code> .

Value

An object of class "scf_logit" and "scf_model_result" with:

results A data frame of pooled estimates (log-odds or odds ratios), standard errors, and test statistics.

fit Model diagnostics including AIC and pseudo-R-Squared (for binomial family).

models List of implicate-level `svyglm` model objects.

call The matched function call.

Details

This function internally calls `scf_glm()` with `family = binomial()` and optionally exponentiates pooled log-odds to odds ratios.

Logistic regression models the probability of a binary outcome using the logit link.

Coefficients reflect the change in log-odds associated with a one-unit change in the predictor.

When `odds = TRUE`, the coefficient estimates and standard errors are transformed from log-odds to odds ratios and approximate SEs.

Warning

When modeling binary outcomes using survey-weighted logistic regression, users may encounter the warning:

```
"non-integer #successes in a binomial glm!"
```

This message is benign. It results from replicate-weighted survey designs where the implied number of "successes" is non-integer. The model is estimated correctly. Coefficients are valid and consistent with maximum likelihood.

For background, see: <https://stackoverflow.com/questions/12953045/warning-non-integer-successes-in-a-binomial-glm-survey-packages>

See Also

`scf_glm()`, `scf_ols()`, `scf_MIcombine()`

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("logit_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
```

```

file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Run logistic regression
model <- scf_logit(scf2022, own ~ age)
summary(model)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)

```

scf_mean

Estimate Mean in Multiply-Imputed SCF Data

Description

Returns the population-level estimate of a continuous variable's weighted mean across the Survey's five implicates. Use this operation to derive an estimate of a population's 'typical' or 'average' score on a continuous variable.

Usage

```
scf_mean(scf, var, by = NULL, verbose = FALSE)
```

Arguments

scf	A <code>scf_mi_survey</code> object created with <code>scf_load()</code> . Must contain five replicate-weighted implicates.
var	A one-sided formula identifying the continuous variable to summarize (e.g., <code>~networth</code>).
by	Optional one-sided formula specifying a discrete grouping variable for stratified means.
verbose	Logical. If TRUE, include implicate-level results in print output. Default is FALSE.

Value

A list of class "scf_mean" with:

results Pooled estimates with standard errors and range across implicates. One row per group, or one row total.

imps A named list of implicate-level estimates.

aux Variable and group metadata.

Details

The mean is a measure of central tendency that represents the arithmetic average of a distribution. It is most appropriate when the distribution is symmetric and not heavily skewed. Unlike the median, the mean is sensitive to extreme values, which may distort interpretation in the presence of outliers. Use this function to describe the "typical" value of a continuous variable in the population or within subgroups.

See Also

[scf_median\(\)](#), [scf_percentile\(\)](#), [scf_xtab\(\)](#), [scf_plot_dist\(\)](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("mean_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Estimate means
scf_mean(scf2022, ~networth)
scf_mean(scf2022, ~networth, by = ~edc1)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_median

Estimate the Population Median of a Continuous SCF Variable

Description

Estimates the median (50th percentile) of a continuous SCF variable. Use this operation to characterize a typical or average value. In contrast to [scf_mean\(\)](#), this function is both uninfluenced by, and insensitive to, outliers.

Usage

```
scf_median(scf, var, by = NULL, verbose = FALSE)
```

Arguments

scf	A <code>scf_mi_survey</code> object created by scf_load() . Must contain five implicates.
var	A one-sided formula specifying the continuous variable of interest (e.g., <code>~networth</code>).
by	Optional one-sided formula for a categorical grouping variable.
verbose	Logical; if TRUE, show implicate-level results.

Value

A list of class "scf_median" with:

results A data frame with pooled medians, standard errors, and range across implicates.

imps A list of implicate-level results.

aux Variable and grouping metadata.

Implementation

This function wraps `scf_percentile()` with $q = 0.5$. The user supplies a `scf_mi_survey` object and a one-sided formula for the variable of interest, with an optional grouping formula. Output includes pooled medians, standard errors, min/max across implicates, and implicate-level values. Point estimates are the mean of the five implicate medians. Standard errors are computed using the Survey of Consumer Finances convention described below, not Rubin's Rules.

Statistical Notes

Median estimates follow the Federal Reserve Board's SCF variance convention. For each implicate, the median is computed with replicate weights via `survey::svyquantile()`. The pooled estimate is the average of the five implicate medians. The pooled variance is $V_{\text{total}} = V_1 + ((m + 1) / m) * B$, where V_1 is the replicate-weight sampling variance from the first implicate and B is the between-implicate variance of the five implicate medians, with $m = 5$ implicates. The reported standard error is $\sqrt{V_{\text{total}}}$. This matches the Federal Reserve Board's published SAS macro for SCF descriptive statistics and is not Rubin's Rules.

See Also

`scf_percentile()`, `scf_mean()`

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("median_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Estimate medians
scf_median(scf2022, ~networth)
scf_median(scf2022, ~networth, by = ~edc1)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_MIcombine

Combine Estimates Across SCF Implicates Using Rubin's Rules

Description

This function implements **Rubin's Rules** for combining multiply-imputed survey model results in the `scf` package. It pools point estimates, variance-covariance matrices, and degrees of freedom across the SCF's five implicates.

Usage

```

scf_MIcombine(results, variances, call = sys.call(), df.complete = Inf)

SE(object, ...)

## S3 method for class 'scf_MIresult'
SE(object, ...)

```

Arguments

<code>results</code>	A list of implicate-level model outputs. Each element must be a named numeric vector or an object with methods for <code>coef()</code> and <code>vcov()</code> . Typically generated internally by modeling functions.
<code>variances</code>	Optional list of variance-covariance matrices. If omitted, extracted using <code>vcov()</code> .
<code>call</code>	Optional. The originating function call. Defaults to <code>sys.call()</code> .
<code>df.complete</code>	Optional degrees of freedom for the complete-data model. Used for small-sample corrections. Defaults to <code>Inf</code> , assuming large-sample asymptotics.
<code>object</code>	A pooled result object of class <code>"scf_MIresult"</code> (for <code>SE()</code>).
<code>...</code>	Not used.

Value

An object of class `"scf_MIresult"` with components:

coefficients Pooled point estimates across implicates.

variance Pooled variance-covariance matrix.

df Degrees of freedom for each parameter, adjusted using Barnard-Rubin formula.

missinfo Estimated fraction of missing information for each parameter.

nimp Number of implicates used in pooling.

call Function call recorded for reproducibility.

Supports `coef()`, [SE\(\)](#), `confint()`, and `summary()` methods.

Scope

`scf_MIcombine()` is used for **model-based analyses** such as [scf_ols\(\)](#), [scf_glm\(\)](#), and [scf_logit\(\)](#), where each implicate's model output includes both parameter estimates and replicate-weighted sampling variances.

Descriptive estimators—functions such as [scf_mean\(\)](#), [scf_percentile\(\)](#), and [scf_median\(\)](#)—do **not** apply Rubin's Rules. They follow the Survey of Consumer Finances convention used in the Federal Reserve Board's SAS macro, combining (i) the replicate-weight sampling variance from implicate 1 with (ii) the between-implicate variance scaled by $(m + 1)/m$.

This separation is intentional: descriptive statistics in `scf` aim to reproduce the Survey of Consumer Finances' published standard errors, whereas model-based functions use Rubin's Rules.

Implementation

`scf_MIcombine()` pools a set of implicate-level estimates and their associated variance-covariance matrices using Rubin's Rules.

This includes:

- Calculation of pooled point estimates
- Total variance from within- and between-imputation components
- Degrees of freedom via Barnard-Rubin method
- Fraction of missing information

Inputs are typically produced by modeling functions such as `scf_ols()`, `scf_glm()`, or `scf_logit()`, which return implicate-level coefficient vectors and variance-covariance matrices.

This function is primarily used internally, but may be called directly by advanced users constructing custom estimation routines from implicate-level results.

Details

The SCF provides five implicates per survey wave, each a plausible version of the population under a specific missing-data model. Analysts conduct the same statistical procedure on each implicate, producing a set of five estimates $Q_1, Q_2, \dots, Q_5$. These are then combined using Rubin's Rules, a procedure to combine results across these implicates with an attempt to account for:

- **Within-imputation variance:** Uncertainty from complex sample design
- **Between-imputation variance:** Uncertainty due to missing data

For a scalar quantity Q , the pooled estimate and total variance are calculated as:

$$\begin{aligned}\bar{Q} &= \frac{1}{M} \sum Q_m \\ \bar{U} &= \frac{1}{M} \sum U_m \\ B &= \frac{1}{M-1} \sum (Q_m - \bar{Q})^2 \\ T &= \bar{U} + \left(1 + \frac{1}{M}\right) B\end{aligned}$$

Where:

- M is the number of implicates (typically 5 for SCF)
- Q_m is the estimate from implicate m
- U_m is the sampling variance of Q_m , accounting for replicate weights and design

The total variance T reflects both within-imputation uncertainty (sampling error) and between-imputation uncertainty (missing-data imputation).

The standard error of the pooled estimate is $\sqrt{T}$. Degrees of freedom are adjusted using the Barnard-Rubin method:

$$\nu = (M-1) \left(1 + \frac{\bar{U}}{(1 + \frac{1}{M})B}\right)^2$$

The fraction of missing information (FMI) is also reported: it reflects the proportion of total variance attributable to imputation uncertainty.

References

Barnard J, Rubin DB. Small-sample degrees of freedom with multiple imputation. doi:10.1093/biomet/86.4.948.

Little RJA, Rubin DB. Statistical analysis with missing data. ISBN: 9780470526798.

U.S. Federal Reserve. Codebook for 2022 Survey of Consumer Finances. <https://www.federalreserve.gov/econres/scfind>

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("MIcombine_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Pool simple survey mean for mock data
outlist <- lapply(scf2022$mi_design, function(d) survey::svymean(~I(age >= 65), d))
pooled <- scf_MIcombine(outlist)      # vcov/coef extracted automatically
SE(pooled); coef(pooled)

unlink(td, recursive = TRUE, force = TRUE)
```

scf_model_result_methods

S3 Methods for scf_model_result Objects

Description

Generic S3 methods dispatched on objects of class "scf_model_result", as returned by [scf_ols](#), [scf_glm](#), [scf_logit](#), and [scf_quantreg](#).

`coef()` Pooled coefficient estimates (Rubin's Rules).

`vcov()` Pooled variance-covariance matrix.

`AIC()` Mean AIC across implicates.

`residuals()` Residuals from the first implicate model (diagnostic use only).

`predict()` Mean predictions pooled across all five implicate models.

`formula()` The model formula.

Usage

```
## S3 method for class 'scf_model_result'
formula(x, ...)

## S3 method for class 'scf_model_result'
residuals(object, ...)

## S3 method for class 'scf_model_result'
```

```

coef(object, ...)

## S3 method for class 'scf_model_result'
vcov(object, ...)

## S3 method for class 'scf_model_result'
AIC(object, k = 2, ...)

## S3 method for class 'scf_model_result'
predict(object, newdata, type = "link", ...)

```

Arguments

<code>x</code>	An object of class "scf_model_result" (for formula).
<code>...</code>	Additional arguments (not used by most methods).
<code>object</code>	An object of class "scf_model_result".
<code>k</code>	Penalty term passed to <code>AIC()</code> (default 2 for AIC, $\log(n)$ for BIC).
<code>newdata</code>	Optional data frame of new observations for <code>predict()</code> . If missing, predictions are made on the original training data.
<code>type</code>	Prediction scale for <code>predict()</code> : "link" (default) or "response".

scf_ols

Estimate an Ordinary Least Squares Regression on SCF Microdata

Description

Computes an OLS regression on SCF data using `svyglm()` across the SCF's five implicates. Returns coefficient estimates, standard errors, test statistics, and model diagnostics.

Usage

```
scf_ols(object, formula)
```

Arguments

<code>object</code>	A <code>scf_mi_survey</code> object created with <code>scf_load()</code> and <code>scf_design()</code> . Must contain five implicates with replicate weights.
<code>formula</code>	A model formula specifying a continuous outcome and predictor variables (e.g., <code>networth ~ income + age</code>).

Details

Fits a replicate-weighted linear regression model to each implicate of multiply-imputed SCF data and pools coefficients and standard errors using Rubin's Rules.

Value

An object of class "scf_ols" and "scf_model_result" with:

results A data frame of pooled coefficients, standard errors, t-values, p-values, and significance stars.

fit A list of model diagnostics including mean AIC, standard deviation of AIC, mean R-squared, and its standard deviation.

imps A list of implicate-level svyglm model objects.

call The matched call used to produce the model.

Implementation

Ordinary least squares (OLS) regression estimates the linear relationship between a continuous outcome and one or more predictor variables. Each coefficient represents the expected change in the outcome for a one-unit increase in the corresponding predictor, holding all other predictors constant.

Use this function to model associations between SCF variables while accounting for complex survey design and multiple imputation.

This function takes a `scf_mi_survey` object and a model formula. Internally, it fits a weighted linear regression to each implicate using `survey::svyglm()`, extracts coefficients and variance-covariance matrices, and pools them via [scf_MIcombine\(\)](#).

See Also

[scf_glm\(\)](#), [scf_logit\(\)](#), [scf_MIcombine\(\)](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("ols_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Run OLS model
model <- scf_ols(scf2022, networth ~ income + age)
summary(model)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_pctile_sum

*Summarize SCF Variables by Percentile Group***Description**

Creates a percentile-based grouping variable for a continuous SCF variable and optionally computes a summary statistic within each group. Two methods are available. The "implicate" method (the default) computes survey-weighted quantile thresholds separately within each implicate, the statistically preferable approach under multiple imputation, as it correctly accounts for between-implicate variation in imputed values. The "stack" method replicates the Federal Reserve's published convention, in which all five implicates are pooled with weights divided by five, a single set of thresholds is computed from the pooled distribution, and a flat weighted statistic is computed directly on the stacked data.

Usage

```
scf_pctile_sum(
  scf,
  var,
  probs = seq(0, 1, by = 0.1),
  labels = NULL,
  varname = NULL,
  method = c("implicate", "stack"),
  stat = c("mean", "median", "none"),
  stat_var = NULL
)

scf_pctile_cut(
  scf,
  var,
  probs = seq(0, 1, by = 0.1),
  labels = NULL,
  varname = NULL,
  method = c("implicate", "stack"),
  stat = c("mean", "median", "none"),
  stat_var = NULL
)
```

Arguments

scf	A scf_mi_survey object created with scf_load .
var	A one-sided formula naming the continuous variable to cut (e.g., ~networth).
probs	Numeric vector with values in between 0 and 1 defining group boundaries, including 0 and 1 as endpoints. Defaults to deciles (seq(0, 1, by = 0.1)).
labels	Optional character vector of group labels, length equal to length(probs) - 1. If NULL (default), labels are generated automatically in the form "p0-p10", "p10-p20", etc.
varname	Optional name for the new grouping variable. Defaults to "{var}_pctile" (e.g., "networth_pctile").

method	Character. One of "implicate" (default) or "stack". See Details.
stat	Character. One of "mean" (default), "median", or "none". When "none", returns the updated <code>scf_mi_survey</code> object with the grouping variable added and no summary statistic computed. When "mean" or "median", computes the statistic within each percentile group and returns a data frame of results.
stat_var	Optional one-sided formula naming the variable to summarize within each group. Defaults to <code>var</code> if not supplied.

Details

The two methods will generally produce similar but not identical results. Use `method = "stack"` when exact replication of the Federal Reserve's published SCF tables is required. Note that `method = "stack"` with `stat != "none"` computes a flat weighted statistic on the pooled stacked data and does not use replicate-weight survey machinery or Rubin's combining rules. No standard error is returned in this case.

When `stat = "none"`, the function returns the input `scf_mi_survey` object with the grouping variable added, ready to pass to `scf_mean`, `scf_median`, or other estimation functions via the `by` argument. For standard grouping variables already published by the Fed, such as net worth percentile (`nwcat`) and income percentile (`inccat`), those variables are included directly in the data returned by `scf_load` and can be passed to `by` without calling `scf_pctile_sum`.

Value

When `stat = "none"`, returns the input `scf_mi_survey` object with a new factor variable added to each `implicate`'s data frame. The variable is named according to `varname` (default: `"{var}_pctile"`), and can be passed directly to `scf_mean`, `scf_median`, or other estimation functions via the `by` argument. When `stat = "mean"` or `"median"` with `method = "implicate"`, returns the output of `scf_mean` or `scf_median` respectively. When `stat != "none"` with `method = "stack"`, returns a data frame with columns `group`, `variable`, and `estimate`. No standard error is returned for the `stack` method.

See Also

[scf_mean](#), [scf_median](#), [scf_percentile](#), [scf_update_by_implicate](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("pctile_sum_")
dir.create(td)
src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Mean net worth, top vs bottom 90 percent, stack method (fast)
scf_pctile_sum(scf2022, ~networth,
  probs = c(0, 0.9, 1),
  labels = c("bottom90", "top10"),
  method = "stack")

## Not run:
# Implicate method (default): requires full SCF data; unreliable on mock data
scf_pctile_sum(scf2022, ~networth)
```

```
# Return grouping variable only, no summary statistic (implicate method)
scf2022 <- scf_pctile_sum(scf2022, ~networth,
                        probs = c(0, 0.9, 1),
                        labels = c("bottom90", "top10"),
                        stat = "none")

## End(Not run)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_percentile

Estimate Percentiles in SCF Microdata

Description

This function estimates a weighted percentile of a continuous variable in the Survey of Consumer Finances (SCF). Two methods are available. The default "implicate" method estimates the percentile separately within each implicate using `survey::svyquantile()` and pools the results following the SCF Bulletin variance convention. The "stack" method replicates the Federal Reserve's published convention by pooling all five implicates with weights divided by five and computing a single weighted quantile from the combined sample. The two methods will generally produce similar but not identical results. Use `method = "stack"` when exact replication of Federal Reserve published figures is required.

Usage

```
scf_percentile(
  scf,
  var,
  q = 0.5,
  by = NULL,
  verbose = FALSE,
  method = c("implicate", "stack")
)
```

Arguments

scf	A <code>scf_mi_survey</code> object created with <code>scf_load()</code> . Must contain the list of replicate-weighted designs for each implicate in <code>scf\$mi_design</code> .
var	A one-sided formula naming the continuous variable to summarize (for example <code>~networth</code>).
q	Numeric percentile in between 0 and 1. Default 0.5 (median).
by	Optional one-sided formula naming a categorical grouping variable. If supplied, the percentile is estimated separately within each group.
verbose	Logical. If TRUE, include implicate-level estimates in the returned object for inspection. Default FALSE.

method Character. One of "implicate" (default) or "stack". The implicate method averages percentile estimates across the five implicates using the SCF Bulletin variance convention. The stack method pools all five implicates into a single dataset with weights divided by five and computes a single weighted quantile, replicating the Federal Reserve's published figures. When method = "stack" no standard error is returned, as the stacked estimate is a deterministic weighted quantile rather than a model-based estimate.

Details

The implicate method computes estimates as follows:

1. For each implicate, estimate the requested percentile using `survey::svyquantile()` with `se = TRUE`.
2. The reported point estimate is the mean of the M implicate-specific percentile estimates.
3. The standard error follows the SCF Bulletin SAS macro convention:

$$V_{\text{total}} = V1 + ((M + 1) / M) * B$$

where:

- V1 is the replicate-weight sampling variance of the percentile from the first implicate only.
- B is the between-implicate variance of the percentile estimates.

The reported standard error is $\sqrt{V_{\text{total}}}$.

4. If a grouping variable is supplied, the same logic is applied separately within each group.

The stack method returns no standard error, as the estimate is a deterministic weighted quantile rather than a model-based estimate.

Value

An object of class "scf_percentile" containing:

results A data frame containing pooled percentile estimates, pooled standard errors, and implicate min/max values. One row per group (if by is supplied) or one row otherwise.

imps A list of implicate-level percentile estimates and standard errors.

aux A list containing the variable name, optional group variable name, and the quantile requested.

verbose Logical flag indicating whether implicate-level estimates should be printed by `print()` or `summary()`.

References

Federal Reserve Board. 2023c. "SAS Macro: Variable Definitions." <https://www.federalreserve.gov/econres/files/bulletin>

See Also

`scf_median()`, `scf_mean()`

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()` for actual SCF data
td <- tempfile("percentile_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Estimate the 75th percentile of net worth
scf_percentile(scf2022, ~networth, q = 0.75)

# Estimate the median net worth by ownership group
scf_percentile(scf2022, ~networth, q = 0.5, by = ~own)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
rm(scf2022)
```

scf_plot_bbar

Stacked Bar Chart of Two Discrete Variables in SCF Data

Description

Visualizes a discrete-discrete bivariate distribution using stacked bars based on pooled cross-tabulations from [scf_xtab\(\)](#). Use this function to visualize the relationship between two discrete variables.

Usage

```
scf_plot_bbar(
  design,
  rowvar,
  colvar,
  scale = c("percent", "count"),
  percent_by = c("total", "row", "col"),
  title = NULL,
  xlab = NULL,
  ylab = NULL,
  fill_colors = NULL,
  row_labels = NULL,
  col_labels = NULL
)
```

Arguments

design	A <code>scf_mi_survey</code> object created by scf_load() . Must contain five implicates with replicate weights.
rowvar	A one-sided formula for the x-axis grouping variable (e.g., <code>~edc1</code>).
colvar	A one-sided formula for the stacked fill variable (e.g., <code>~race1</code>).

scale	Character. One of "percent" (default) or "count".
percent_by	Character. One of "total" (default), "row", or "col" — determines normalization base when scale = "percent".
title	Optional character string for the plot title.
xlab	Optional character string for the x-axis label.
ylab	Optional character string for the y-axis label.
fill_colors	Optional vector of fill colors to pass to <code>ggplot2::scale_fill_manual()</code> .
row_labels	Optional named vector to relabel row categories (x-axis).
col_labels	Optional named vector to relabel col categories (legend).

Value

A `ggplot2` object.

Implementation

This function calls `scf_xtab()` to estimate the joint distribution of two categorical variables across multiply-imputed SCF data. The result is translated into a `ggplot2` stacked bar chart using pooled counts or normalized percentages.

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("plot_bbar_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Stacked bar chart: education by ownership
scf_plot_bbar(scf2022, ~own, ~edcl)

# Example for real analysis: Column percentages instead of total percent
scf_plot_bbar(scf2022, ~own, ~edcl, percent_by = "col")

# Example for real analysis: Raw counts (estimated number of households)
scf_plot_bbar(scf2022, ~own, ~edcl, scale = "count")

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_plot_cbar

Bar Plot of Summary Statistics by Grouping Variable in SCF Data

Description

Computes and plots a grouped summary statistic (either a mean, median, or quantile) for a continuous variable across a discrete factor. Estimates are pooled across implicates using `scf_mean()`, `scf_median()`, or `scf_percentile()`. Use this function to visualize the bivariate relationship between a discrete and a continuous variable.

Usage

```
scf_plot_cbar(
  design,
  yvar,
  xvar,
  stat = "mean",
  title = NULL,
  xlab = NULL,
  ylab = NULL,
  fill = "#0072B2",
  angle = 30,
  label_map = NULL
)
```

Arguments

design	A <code>scf_mi_survey</code> object from <code>scf_load()</code> .
yvar	One-sided formula for the continuous variable (e.g., <code>~networth</code>).
xvar	One-sided formula for the grouping variable (e.g., <code>~racecl</code>).
stat	"mean" (default), "median", or a quantile (numeric between 0 and 1).
title	Plot title (optional).
xlab	X-axis label (optional).
ylab	Y-axis label (optional).
fill	Bar fill color. Default is "#0072B2".
angle	Angle of x-axis labels. Default is 30.
label_map	Optional named vector to relabel x-axis category labels.

Value

A `ggplot2` object.

Implementation

The user specifies a continuous outcome (`yvar`) and a discrete grouping variable (`xvar`) via one-sided formulas. Group means are plotted by default. Medians or other percentiles can be specified via the `stat` argument.

Results are plotted using `ggplot2::geom_col()`, styled with `scf_theme()`, and optionally customized with additional arguments (e.g., axis labels, color, angles).

See Also

[scf_mean\(\)](#), [scf_median\(\)](#), [scf_percentile\(\)](#), [scf_theme\(\)](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("plot_cbar_")
dir.create(td)
```

```
src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Plot mean net worth by education level
scf_plot_dbar(scf2022, ~networth, ~edcl, stat = "mean")

# Example for real analysis: Visualize 90th percentile of income by education
scf_plot_dbar(scf2022, ~income, ~edcl, stat = 0.9, fill = "#D55E00")

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_plot_dbar

*Plot Bar Chart of a Discrete Variable from SCF Data***Description**

Creates a bar chart that visualizes the distribution of a discrete variable.

Usage

```
scf_plot_dbar(
  design,
  variable,
  title = NULL,
  xlab = NULL,
  ylab = "Percent",
  angle = 30,
  fill = "#0072B2",
  label_map = NULL
)
```

Arguments

design	A scf_mi_survey object created by scf_load() . Must contain valid implicates.
variable	A one-sided formula specifying a categorical variable (e.g., ~racecl).
title	Optional character string for the plot title. Default: "Distribution of <variable>".
xlab	Optional x-axis label. Default: variable name.
ylab	Optional y-axis label. Default: "Percent".
angle	Integer. Rotation angle for x-axis labels. Default is 30.
fill	Fill color for bars. Default is "#0072B2".
label_map	Optional named vector to relabel x-axis category labels.

Value

A ggplot2 object representing the pooled bar chart.

Implementation

This function internally calls `scf_freq()` to compute population proportion estimates, which are then plotted using `ggplot2::geom_col()`. The default output is scaled to percent and can be customized via title, axis labels, angle, and color.

Details

Produces a bar chart of category proportions from a one-way tabulation, pooled across SCF implicates using `scf_freq()`. This function summarizes weighted sample composition and communicates categorical distributions effectively in descriptive analysis.

Dependencies

Requires the `ggplot2` package.

See Also

`scf_freq()`, `scf_plot_bbar()`, `scf_xtab()`

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("plot_dbar_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Bar chart of education categories
scf_plot_dbar(scf2022, ~edcl)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_plot_dist

Plot a Univariate Distribution of an SCF Variable

Description

This function provides a unified plotting interface for visualizing the distribution of a single variable from multiply-imputed SCF data. Discrete variables produce bar charts of pooled proportions; continuous variables produce binned histograms. Use this function to visualize the univariate distribution of an SCF variable.

Usage

```
scf_plot_dist(
  design,
  variable,
  bins = 30,
  title = NULL,
  xlab = NULL,
  ylab = "Percent",
  angle = 30,
  fill = "#0072B2",
  labels = NULL
)
```

Arguments

<code>design</code>	A <code>scf_mi_survey</code> object created by <code>scf_load()</code> .
<code>variable</code>	A one-sided formula specifying the variable to plot.
<code>bins</code>	Number of bins for continuous variables. Default is 30.
<code>title</code>	Optional plot title.
<code>xlab</code>	Optional x-axis label.
<code>ylab</code>	Optional y-axis label. Default is "Percent".
<code>angle</code>	Angle for x-axis tick labels. Default is 30.
<code>fill</code>	Fill color for bars. Default is "#0072B2".
<code>labels</code>	Optional named vector of custom axis labels (for discrete variables only).

Value

A `ggplot2` object.

Implementation

For discrete variables (factor or numeric with ≤ 25 unique values), the function uses `scf_freq()` to calculate category proportions and produces a bar chart. For continuous variables, it bins values across implicates and estimates Rubin-pooled frequencies for each bin.

Users may supply a named vector of custom axis labels using the `labels` argument.

See Also

[scf_theme\(\)](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("plot_dist_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)
```

```
# Example for real analysis: Distribution of homeownership
scf_plot_dist(scf2022, ~own)

# Example for real analysis: Distribution of age
scf_plot_dist(scf2022, ~age, bins = 10)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_plot_hex

*Hexbin Plot of Two Continuous SCF Variables***Description**

Visualizes the bivariate relationship between two continuous SCF variables using hexagonal bins.

Usage

```
scf_plot_hex(design, x, y, bins = 50, title = NULL, xlab = NULL, ylab = NULL)
```

Arguments

design	A scf_mi_survey object created by <code>scf_load()</code> .
x	A one-sided formula for the x-axis variable (e.g., <code>~income</code>).
y	A one-sided formula for the y-axis variable (e.g., <code>~networth</code>).
bins	Integer. Number of hexagonal bins along the x-axis. Default is 50.
title	Optional character string for the plot title.
xlab	Optional x-axis label. Defaults to the variable name.
ylab	Optional y-axis label. Defaults to the variable name.

Value

A ggplot2 object displaying a Rubin-pooled hexbin plot.

Implementation

The function stacks all implicates into one data frame, retains replicate weights, and uses `ggplot2::geom_hex()` to produce a density-style scatterplot. The color intensity of each hexagon reflects the Rubin-pooled weighted count of households in that cell. Missing values are excluded.

This plot is especially useful for visualizing joint distributions with large samples and skewed marginals, such as net worth vs. income.

Aesthetic Guidance

This plot uses a log-scale fill and viridis palette to highlight variation in density. To adjust the visual style globally, use `scf_theme()` or set it explicitly with `ggplot2::theme_set(scf_theme())`. For mobile-friendly or publication-ready appearance, export the plot at 5.5 x 5.5 inches, 300 dpi.

Dependencies

Requires the `ggplot2` package. The fill scale uses `scale_fill_viridis_c()` from `ggplot2`. Requires the **hexbin** package. The function will stop with an error if it is not installed.

See Also

[scf_corr\(\)](#), [scf_plot_smooth\(\)](#), [scf_theme\(\)](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("plot_hex_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Plot hexbin of income vs. net worth
# Note: mock data has ~75 rows per implicate; hexbin output will be sparse.
# Results on full SCF data will show a meaningful joint density.
scf_plot_hex(scf2022, ~income, ~networth)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_plot_hist

Histogram of a Continuous Variable in Multiply-Imputed SCF Data

Description

Produces a histogram of a continuous SCF variable by binning across implicates, pooling weighted bin counts using [scf_freq\(\)](#), and plotting the result. Values outside `xlim` are clamped into the nearest endpoint to ensure all observations are included and replicate-weighted bins remain stable.

Usage

```
scf_plot_hist(
  design,
  variable,
  bins = 30,
  xlim = NULL,
  title = NULL,
  xlab = NULL,
  ylab = "Weighted Count",
  fill = "#0072B2"
)
```

Arguments

design	A scf_mi_survey object from scf_load() .
variable	A one-sided formula indicating the numeric variable to plot.
bins	Number of bins (default: 30).
xlim	Optional numeric range. Values outside will be included in edge bins.
title	Optional plot title.
xlab	Optional x-axis label. Defaults to the variable name.
ylab	Optional y-axis label. Defaults to "Weighted Count".
fill	Fill color for bars (default: "#0072B2").

Value

A ggplot2 object representing the Rubin-pooled histogram.

Implementation

This function bins a continuous variable (after clamping to xlim if supplied), applies the same `cut()` breaks across implicates using [scf_update_by_implicate\(\)](#), and computes Rubin-pooled frequencies with [scf_freq\(\)](#). Results are filtered to remove bins with undefined proportions and then plotted using `ggplot2::geom_col()`.

The logic here is specific to operations where the bin assignment must be computed **within** each implicate, not after pooling. This approach ensures consistent binning and stable pooled estimation in the presence of multiply-imputed microdata.

See Also

[scf_freq\(\)](#), [scf_plot_dbar\(\)](#), [scf_plot_smooth\(\)](#), [scf_update_by_implicate\(\)](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("plot_hist_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Plot histogram of age
scf_plot_hist(scf2022, ~age, bins = 10)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_plot_smooth

*Smoothed Distribution Plot of a Continuous Variable in SCF Data***Description**

Draws a smoothed distribution plot of a continuous variable in the SCF. Use this function to visualize a single continuous variable's distribution.

Usage

```
scf_plot_smooth(
  design,
  variable,
  binwidth = NULL,
  xlim = NULL,
  method = "loess",
  span = 0.2,
  color = "blue",
  xlab = NULL,
  ylab = "Percent of Households",
  title = NULL
)
```

Arguments

design	A scf_mi_survey object created by <code>scf_load()</code> .
variable	A one-sided formula specifying a continuous variable (e.g., <code>~networth</code>).
binwidth	Optional bin width. Default uses Freedman–Diaconis rule.
xlim	Optional numeric vector of length 2 to truncate axis.
method	Character. Smoothing method: "loess" (default) or "lm".
span	Numeric LOESS span. Default is 0.2. Ignored if method = "lm".
color	Line color. Default is "blue".
xlab	Optional label for x-axis. Defaults to the variable name.
ylab	Optional label for y-axis. Defaults to "Percent of Households".
title	Optional plot title.

Value

A ggplot2 object.

Implementation

Visualizes the weighted distribution of a continuous SCF variable by stacking implicates, binning observations, and smoothing pooled proportions. This function is useful for examining distribution shape, skew, or modality in variables like income or wealth.

All implicates are stacked and weighted, binned across a data-driven or user-specified bin width. Each bin's weight share is calculated, and a smoothing curve is fit to the resulting pseudo-density.

See Also

[scf_plot_hist\(\)](#), [scf_plot_dist\(\)](#), [scf_theme\(\)](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("plot_smooth_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Plot smoothed distribution
scf_plot_smooth(scf2022, ~networth, xlim = c(0, 2e6),
               method = "loess", span = 0.25)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_prop_test

Test a Proportion in SCF Data

Description

Tests a binary variable's proportion against a null hypothesis (one-sample), or compares proportions across two groups (two-sample). Supports two-sided, less-than, or greater-than alternatives.

Usage

```
scf_prop_test(
  design,
  var,
  group = NULL,
  p = 0.5,
  alternative = c("two.sided", "less", "greater"),
  conf.level = 0.95
)
```

Arguments

design	A <code>scf_mi_survey</code> object created by scf_load() . Must contain replicate-weighted implicates.
var	A one-sided formula indicating a binary variable (e.g., <code>~rich</code>).
group	Optional one-sided formula indicating a binary grouping variable (e.g., <code>~female</code>). If omitted, a one-sample test is performed.
p	Null hypothesis value. Defaults to 0.5 for one-sample, 0 for two-sample tests.
alternative	Character. One of "two.sided" (default), "less", or "greater".
conf.level	Confidence level for the confidence interval. Default is 0.95.

Value

An object of class "scf_prop_test" with:

results A data frame with the pooled estimate, standard error, z-statistic, p-value, confidence interval, and significance stars.

proportions (Only in two-sample tests) A data frame of pooled proportions by group.

fit A list describing the method, null value, alternative hypothesis, and confidence level.

Statistical Notes

Proportions are computed in each implicate using weighted means, and variances are approximated under the binomial model. Rubin's Rules are applied to pool point estimates and standard errors. For pooling details, see [scf_MIcombine\(\)](#).

See Also

[scf_ttest\(\)](#), [scf_mean\(\)](#), [scf_MIcombine\(\)](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("proptest_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Wrangle data for example
scf2022 <- scf_update(scf2022,
  rich = network > 1e6,
  female = factor(hhsex, levels = 1:2, labels = c("Male", "Female")),
  over50 = age > 50
)

# Example for real analysis: One-sample test
scf_prop_test(scf2022, ~rich, p = 0.10)

# Example for real analysis: Two-sample test
scf_prop_test(scf2022, ~rich, ~female, alternative = "less")

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

Description

`scf_quantreg()` estimates a linear quantile regression at a user-specified quantile (τ) across the five SCF implicates. Each implicate is fit independently via `quantreg::rq()` using the SCF final sampling weights. Coefficient estimates and variance-covariance matrices are then pooled across implicates using `scf_MIcombine()`.

Unlike `scf_ols()`, which models the conditional mean, quantile regression models the conditional quantile of the outcome distribution. This makes it especially useful for analyzing wealth and income data, which are highly right-skewed: the conditional median ($\tau = 0.5$) and upper quantiles ($\tau = 0.75$, $\tau = 0.90$) describe the distribution more completely than the mean alone.

Usage

```
scf_quantreg(
  object,
  formula,
  tau = 0.5,
  se = c("nid", "iid", "ker", "boot", "replicate"),
  ...
)
```

Arguments

<code>object</code>	A <code>scf_mi_survey</code> object created with <code>scf_load()</code> or <code>scf_design()</code> .
<code>formula</code>	A model formula specifying the outcome and predictors, e.g., <code>networth ~ age + factor(edc1)</code> . The outcome should be numeric.
<code>tau</code>	Numeric scalar in (0, 1) specifying the quantile to estimate. Defaults to 0.5 (median regression). To estimate multiple quantiles, call <code>scf_quantreg()</code> separately for each. Analytical SE methods ("nid", "iid", "ker", "boot") may be unreliable at quantiles with high mass points; prefer <code>se = "replicate"</code> in such cases.
<code>se</code>	Character string specifying the standard error estimation method for within-implicate variance. One of "nid" (default), "iid", "ker", "boot", or "replicate". See the Standard Error Methods section for details. "replicate" is theoretically preferred for quantile regression but is computationally intensive.
<code>...</code>	Additional arguments passed to <code>quantreg::rq()</code> .

Details

Fits a survey-weighted quantile regression to each implicate of multiply-imputed SCF data and pools coefficients and standard errors using Rubin's Rules.

Value

An object of class "scf_quantreg" and "scf_model_result" with:

- `results` A data frame of pooled coefficients, standard errors, t-values, p-values, and significance stars.
- `tau` The quantile estimated.
- `se_method` The SE method used.
- `fit` A list of goodness-of-fit statistics. See the **Goodness of Fit** section for details. Components: `rho`, `rho_null`, `r1`, `r1_adj`, `nobs`, `nobs_mean`.

`models` A list of implicate-level rq model objects for direct inspection.

`fit_errors` Character vector of any implicate-level errors.

`call` The matched call.

`formula` The model formula.

Standard Error Methods

The `se` argument controls how within-implicate variance is estimated. All methods produce a variance-covariance matrix passed to `scf_MIcombine()`.

- "replicate" (recommended)** Replication-based variance estimation. For each implicate, the model is re-fit using each of the SCF's 999 replicate weight vectors. Variance is accumulated as the weighted sum of squared deviations from the full-weight estimate, matching the SCF's own published variance methodology via `survey::withReplicates()`. This method is theoretically preferred for quantile regression because replication-based variance estimators are consistent for sample quantiles, whereas analytical (sandwich) estimators are not guaranteed consistent for nonsmooth statistics (Rust and Rao, 1996). Computationally intensive (~5,000 model fits for five implicates).
- "nid" (default)** Non-iid sandwich estimator. Allows the conditional sparsity (density of the error distribution at the quantile) to vary across observations. Appropriate when the shape of the error distribution differs across the covariate space, which is typical for skewed outcomes such as wealth and income. Unreliable near quantiles with high mass points (e.g., $\tau \leq 0.25$ when net worth has substantial mass at zero); use `se = "replicate"` in such cases.
- "iid"** Assumes identically distributed errors, i.e., constant sparsity across all observations. Implements the covariance formula from Koenker and Bassett (1978, Theorem 4.2): $[\theta(1 - \theta)/f(\xi(\theta))^2]Q^{-1}$, where $f(\xi(\theta))$ is the density of the error distribution at its θ -quantile and $Q = \lim T^{-1}X'X$. Fastest analytical option; subject to the same mass-point caveat as "nid".
- "ker"** Kernel smoothing estimate of the conditional sparsity. More data-adaptive than "nid" but slower. Suitable for large samples.
- "boot"** Pairs bootstrap over observations. Provides distribution-free variance estimates but is the slowest analytical option. Useful for robustness checks.

Goodness of Fit

The `fit` component of the returned object contains per-quantile goodness-of-fit measures following Koenker and Machado (1999). These are computed by comparing the full model to an intercept-only null model at the same τ .

- `rho` Mean minimized weighted sum of absolute residuals ($\sum \rho_\tau(y_i - x'_i\hat{\beta})$) from the full model, averaged across implicates. This is the quantile regression analog of the residual sum of squares.
- `rho_null` Same quantity from the intercept-only null model. Larger values relative to `rho` indicate greater explanatory power.
- `r1` The Koenker-Machado $R^1(\tau)$ statistic: $1 - \bar{V}(\tau)/\tilde{V}(\tau)$, where $\bar{V}$ is the mean full-model objective and $\tilde{V}$ is the mean null-model objective, each averaged across implicates. Ranges from 0 to 1. Measures the proportional reduction in the weighted sum of absolute residuals due to the covariates at quantile τ . This is a *local* measure for the specific quantile estimated; it is not a global summary of fit across the distribution.

`r1_adj` Degrees-of-freedom-adjusted $R^1(\tau)$: $1 - (1 - R^1) \cdot n / (n - p)$, where n is the mean number of observations and p is the number of estimated parameters. Penalizes model complexity analogously to adjusted R^2 in OLS. Note: this adjustment is not derived from the asymptotic theory in Koenker and Machado (1999) and should be interpreted descriptively.

`nobs` Integer vector of per-implicate sample sizes.

`nobs_mean` Mean sample size across successful implicates.

Implementation

For each implicate:

1. Extracts the survey data and final sampling weights from the `svyrep.design` object.
2. Fits `quantreg::rq()` at quantile `tau` with the final weights.
3. Estimates the within-implicate variance-covariance matrix using the method specified by `se`.
4. Fits an intercept-only `quantreg::rq()` at the same `tau` and weights to obtain the null-model objective value for goodness-of-fit.

SCF public-use microdata contain no missing values; each implicate is a complete dataset. Pooled estimates follow Rubin's Rules (see `scf_MIcombine()`); the total variance combines within-implicate sampling uncertainty and between-implicate imputation uncertainty.

References

- Koenker R, Bassett G. Regression quantiles. *Econometrica*. 1978;46(1):33–50. doi:10.2307/1913643
- Koenker R, Machado JAF. Goodness of fit and related inference processes for quantile regression. *Journal of the American Statistical Association*. 1999;94(448):1296–1310. doi:10.2307/2669943
- Rust KF, Rao JNK. Variance estimation for complex surveys using replication techniques. *Statistical Methods in Medical Research*. 1996;5(3):283–310. doi:10.1177/096228029600500305

See Also

`scf_ols()`, `scf_glm()`, `scf_MIcombine()`, `quantreg::rq()`

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("qreg-")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: median regression of net worth on age and education
m_med <- scf_quantreg(scf2022, networth ~ age + factor(edc1), tau = 0.5)
summary(m_med)

# Access goodness-of-fit statistics
m_med$fit$r1
m_med$fit$r1_adj
```

```
# Example for real analysis: 75th-percentile regression
m_75 <- scf_quantreg(scf2022, networth ~ age + factor(edcl), tau = 0.75)
summary(m_75)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_regtable	<i>Format and Display Regression Results from Multiply-Imputed SCF Models</i>
--------------	---

Description

This function formats and aligns coefficient estimates, standard errors, and significance stars from one or more SCF regression model objects (e.g., from `scf_ols()`, `scf_logit()`, `scf_quantreg()`, or `scf_glm()`).

Usage

```
scf_regtable(
  ...,
  model.names = NULL,
  digits = 0,
  auto_digits = FALSE,
  labels = NULL,
  output = c("console", "markdown", "latex", "csv"),
  file = NULL
)
```

Arguments

<code>...</code>	One or more SCF regression model objects, or a single list of such models.
<code>model.names</code>	Optional character vector naming the models. Defaults to "Model 1", "Model 2", etc.
<code>digits</code>	Integer specifying decimal places for numeric formatting when <code>auto_digits = FALSE</code> . Default is 0.
<code>auto_digits</code>	Logical; if TRUE, uses adaptive decimal places: 0 digits for large numbers (≥ 1000), 2 digits for moderate (≥ 1), and 3 digits for smaller values.
<code>labels</code>	Optional named character vector or labeling function to replace term names with descriptive labels.
<code>output</code>	Output format: one of "console" (print to console), "markdown" (print Markdown table for R Markdown), "latex" (print LaTeX table for PDF compilation), or "csv" (write CSV file).
<code>file</code>	File path for CSV output; required if <code>output = "csv"</code> .

Details

It compiles a side-by-side table with terms matched across models, appends model fit statistics (sample size N, R-squared or pseudo-R-squared, quantile tau, and AIC where applicable), and outputs the results as console text, Markdown for R Markdown documents, LaTeX for PDF compilation, or a CSV file.

The function aligns all unique coefficient terms across provided models, formats coefficients with significance stars and standard errors, appends model fit statistics as additional rows, and renders output in the specified format.

Fit statistics rows are automatically selected based on model class:

All models Sample size (N)

OLS models R-squared and AIC

Logit/GLM models Pseudo-R-squared and AIC

Quantile regression Quantile tau, R1(tau), and adjusted R1(tau)

It avoids external dependencies by using base R formatting and simple text, Markdown, LaTeX, or CSV output.

Value

Invisibly returns a data frame with formatted regression results and fit statistics.

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("regtable_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Wrangle data for example: Perform OLS regression
m1 <- scf_ols(scf2022, income ~ age)

# Example for real analysis: Print regression results as a console table
scf_regtable(m1, digits = 2)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_subset

Subset an scf_mi_survey Object

Description

Subsetting refers to the process of retaining only those observations that satisfy a logical (TRUE/FALSE) condition. This function applies such a filter independently to each implicate in an `scf_mi_survey` object created by `scf_design()` via `scf_load()`. The result is a new multiply-imputed, replicate-weighted survey object with appropriately restricted designs.

Usage

```
scf_subset(scf, expr)
```

Arguments

scf	A scf_mi_survey object, typically created by scf_load() .
expr	A logical expression used to filter rows, evaluated separately in each implicate's variable frame (e.g., <code>age < 65 & own == 1</code>).

Value

A new scf_mi_survey object (see [scf_design\(\)](#))

Implementation

Use `scf_subset()` to focus analysis on analytically meaningful sub-populations. For example, to analyze only households headed by seniors:

```
scf2022_seniors <- scf_subset(scf2022, age >= 65)
```

This is especially useful when analyzing populations such as renters, homeowners, specific age brackets, or any group defined by logical expressions over SCF variables.

Details

Filtering is conducted separately in each implicate. This preserves valid design structure but means that the same household may fall into or out of the subset depending on imputed values. For example, a household with five different age imputations—say, 64, 66, 63, 65, and 67—would be classified as a senior in only three of five implicates if subsetting on `age >= 65`.

Empty subsets in any implicate can cause downstream analysis to fail. Always check subgroup sizes after subsetting.

See Also

[scf_load\(\)](#), [scf_update\(\)](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("subset_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Filter for working-age households with positive net worth
scf_sub <- scf_subset(scf2022, age < 65 & networth > 0)
scf_mean(scf_sub, ~income)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

scf_ttest

*T-Test of Means using SCF Microdata***Description**

Tests whether the mean of a continuous variable differs from a specified value (one-sample), or whether group means differ across a binary factor (two-sample). Estimates and standard errors are computed using `svymean()` within each implicate, then pooled using Rubin's Rules. Use this function to test hypotheses about means in the SCF microdata.

Usage

```
scf_ttest(
  design,
  var,
  group = NULL,
  mu = 0,
  alternative = c("two.sided", "less", "greater"),
  conf.level = 0.95
)
```

Arguments

<code>design</code>	A <code>scf_mi_survey</code> object created by <code>scf_load()</code> .
<code>var</code>	A one-sided formula specifying a numeric variable (e.g., <code>~income</code>).
<code>group</code>	Optional one-sided formula specifying a binary grouping variable (e.g., <code>~female</code>).
<code>mu</code>	Numeric. Null hypothesis value. Default is 0.
<code>alternative</code>	Character. One of "two.sided" (default), "less", or "greater".
<code>conf.level</code>	Confidence level for the confidence interval. Default is 0.95.

Value

An object of class `scf_ttest` with:

results A data frame with pooled estimate, standard error, t-statistic, degrees of freedom, p-value, and confidence interval.

means Group-specific means (for two-sample tests only).

fit List describing the test type, null hypothesis, confidence level, and alternative.

See Also

[scf_prop_test\(\)](#), [scf_mean\(\)](#), [scf_MIcombine\(\)](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("ttest_")
dir.create(td)
```

```

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Wrangle data for example: Derive analysis vars
scf2022 <- scf_update(scf2022,
  female = factor(hhsex, levels = 1:2, labels = c("Male", "Female")),
  over50 = age > 50
)

# Example for real analysis: One-sample t-test
scf_ttest(scf2022, ~income, mu = 75000)

# Example for real analysis: Two-sample t-test
scf_ttest(scf2022, ~income, group = ~female)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)

```

scf_update

Create or Alter SCF Variables

Description

Use this function to create or alter SCF variables once the raw data set has been loaded into memory using the `scf_load()` function. This function updates an `scf_mi_survey` object by evaluating transformations within each implicate, and then returning a new object with the new or amended variables.

Most of the time, you can use `scf_update()` to define variables based on simple logical conditions, arithmetic transformations, or categorical binning. These rules are evaluated separately in each implicate, using the same formula. However, if the transformation you want to apply depends on the distribution of the data within each implicate, such as computing an average percentile or ranking households across all implicates, this function will not suffice. In those cases, use [scf_update_by_implicate\(\)](#) to write a custom function that operates on each implicate individually.

Usage

```
scf_update(object, ...)
```

Arguments

<code>object</code>	A <code>scf_mi_survey</code> object, typically created by scf_load() .
<code>...</code>	Named expressions assigning new or modified variables using <code>=</code> syntax. Each expression must return a vector of the same length as the implicate data frame.

Value

The input `scf_mi_survey` object with `mi_design` updated to reflect the new or modified variables. All other attributes (`year`, `n_households`, `mock`) are preserved unchanged.

Usage

Use `scf_update()` during data wrangling to clean, create, or alter variables before calculating statistics or running models. The function is useful when the analyst wishes to:

- Recode missing values that are coded as numeric data
- Recast variables that are not in the desired format (e.g., converting a numeric variable to a factor)
- Create new variables based on existing ones (e.g., calculating ratios, differences, or indicators)

See Also

`scf_load()`, `scf_update_by_implicate()`, `survey::svrepdesign()`

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("update_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Create a binary indicator for being over age 50
scf2022 <- scf_update(scf2022,
  over50 = age > 50
)

# Example: Create a log-transformed income variable
scf2022 <- scf_update(scf2022,
  log_income = log(income + 1)
)

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

`scf_update_by_implicate`

Modify Each Implicate Individually in SCF Data

Description

Each household in SCF data is represented by five *implicates*, which reflect uncertainty from the imputation process. Most transformations — such as computing log income or assigning categorical bins — can be applied uniformly across implicates using `scf_update()`. However, some operations depend on the *internal distribution* of variables within each implicate. For those, you need to modify each one separately.

This function extracts each implicate from the replicate-weighted survey design, applies your transformation, and rebuilds the survey design objects accordingly.

Usage

```
scf_update_by_implicate(object, f)
```

Arguments

object	A <code>scf_mi_survey</code> object from <code>scf_load()</code> .
f	A function that takes one implicate's data frame as its sole argument and returns a modified data frame with the same number of rows. The function signature should be <code>function(df) { ... }</code> . The returned data frame is used to rebuild the replicate-weighted survey design.

Details

Applies a user-defined transformation to each implicate's data frame separately. This is useful when you need to compute values that depend on the distribution within each implicate — such as ranks, percentiles, or groupwise comparisons — which cannot be computed reliably using `scf_update()`.

Value

A modified `scf_mi_survey` object with updated implicate-level designs.

Use this When

- You need implicate-specific quantiles (e.g., flag households in the top 10% of wealth)
- You want to assign percentile ranks (e.g., income percentile by implicate)
- You are computing statistics within groups (e.g., groupwise z-scores)
- You need to derive a variable based on implicate-specific thresholds or bins

See Also

[scf_update\(\)](#)

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("update_by_implicate_")
dir.create(td)
src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: flag households in the top 10% of net worth,
# using the unweighted implicate-specific 90th percentile as the threshold.
scf2022 <- scf_update_by_implicate(scf2022, function(df) {
  threshold <- stats::quantile(df$networth, probs = 0.90, na.rm = TRUE)
  df$top10nw <- df$networth >= threshold
  df
})

# Example for real analysis: compute implicate-specific z-scores of income
scf2022 <- scf_update_by_implicate(scf2022, function(df) {
  mu <- mean(df$income, na.rm = TRUE)
  sigma <- stats::sd(df$income, na.rm = TRUE)
```

```

    df$z_income <- (df$income - mu) / sigma
    df
  })

  # Verify new variable exists
  head(scf2022$mi_design[[1]]$variables$z_income)

  # Do not implement these lines in real analysis: Cleanup for package check
  unlink(td, recursive = TRUE, force = TRUE)

```

scf_xtab

Cross-Tabulate Two Discrete Variables in Multiply-Imputed SCF Data

Description

Computes replicate-weighted two-way cross-tabulations of two discrete variables using multiply-imputed SCF data. Estimates cell proportions and standard errors, with optional scaling of proportions by cell, row, or column. Results are pooled across implicates using Rubin's Rules.

Usage

```
scf_xtab(scf, rowvar, colvar, scale = "cell")
```

Arguments

scf	A <code>scf_mi_survey</code> object, typically created by <code>scf_load()</code> . Must include five implicates with replicate weights.
rowvar	A one-sided formula specifying the row variable (e.g., <code>~edcl</code>).
colvar	A one-sided formula specifying the column variable (e.g., <code>~racecl</code>).
scale	Character. Proportion basis: "cell" (default), "row", or "col".

Value

A list of class "scf_xtab" with:

results Data frame with one row per cell. Columns: row, col, prop, se, row_share, col_share, rowvar, and colvar.

matrices List of matrices: cell (default proportions), row, col, and se.

imps List of implicate-level cell count tables.

aux List with rowvar and colvar names.

Statistical Notes

Implicate-level tables are created using `svytable()` on replicate-weighted designs. Proportions are calculated as shares of total population estimates. Variance across implicates is used to estimate uncertainty. Rubin's Rules are applied in simplified form.

For technical details on pooling logic, see `scf_MIcombine()` or the SCF package manual.

Examples

```
# Do not implement these lines in real analysis:
# Use functions `scf_download()` and `scf_load()`
td <- tempfile("xtab_")
dir.create(td)

src <- system.file("extdata", "scf2022_mock_raw.rds", package = "scf")
file.copy(src, file.path(td, "scf2022.rds"), overwrite = TRUE)
scf2022 <- scf_load(2022, data_directory = td)

# Example for real analysis: Cross-tabulate ownership by education
scf_xtab(scf2022, ~own, ~hhsex, scale = "row")

# Do not implement these lines in real analysis: Cleanup for package check
unlink(td, recursive = TRUE, force = TRUE)
```

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